Drug Summary

Elacestrant (Orserdu, RAD1901) is an oral, non-steroidal selective estrogen receptor degrader (SERD) approved for postmenopausal women and adult men with ER-positive, HER2-negative, *ESR1-mutated* advanced or metastatic breast cancer whose disease has progressed after at least one line of endocrine therapy (all EMERALD patients had prior CDK4/6 inhibitor). It competitively antagonizes ERα (ESR1) and promotes receptor degradation, thereby suppressing estrogen-dependent transcriptional programs that drive luminal (HR+/HER2–) breast cancer. Preclinical work showed potent activity in ER+ patient-derived xenografts, including models resistant to aromatase inhibitors, fulvestrant and CDK4/6 inhibitors.

The phase III EMERALD trial (NCT03778931) randomized patients with ER+/HER2– metastatic breast cancer after prior CDK4/6 inhibitor to elacestrant vs physician's-choice endocrine therapy (fulvestrant or AI). Elacestrant improved progression-free survival (PFS) in the overall population (HR≈0.70) and more markedly in tumors with *ESR1* mutations (HR≈0.55; median PFS ~3.8 vs 1.9 months). These data underpinned FDA and EMA approvals, and elacestrant is now incorporated into NCCN, ESMO and ASCO guidance as a preferred endocrine option for HR+/HER2–, *ESR1-mutated* disease after prior endocrine therapy + CDK4/6 inhibitor. No role is currently established in HER2-positive or triple-negative breast cancer.

Identifiers & Synonyms

ChEMBL ID

- **CHEMBL4297509** (elacestrant)

DrugBank ID

- **DB06374** (elacestrant)

Synonyms / Brand / Codes

- **Generic name:** elacestrant
- **Brand/trade name:** Orserdu™ (US/EU labeling)
- **Development / research codes:** RAD1901, ER-306323 (commonly used in preclinical literature)
- **Chemical class/description:** non-steroidal, orally bioavailable SERD based on a tetrahydronaphthalene core with an ethylamine side chain.

Mechanism of Action (Breast Cancer Context)

Elacestrant is a high-affinity ERα (ESR1) antagonist and degrader. In ER+ breast-cancer cells, RAD1901 (elacestrant) binds ERα, induces a conformational change that disrupts coactivator binding, and targets the receptor for ubiquitin-proteasome–dependent degradation, reducing ER protein levels and estrogen-responsive transcription. This mirrors fulvestrant's biological effect but with an oral, non-steroidal scaffold and favorable pharmacokinetics (achievable systemic exposures at recommended dose). The FDA label explicitly classifies elacestrant as a SERD and attributes its antitumor activity to ER antagonism and degradation.

In luminal (ER+/HER2–) breast cancer, ERα governs proliferation and survival by regulating estrogen response gene sets, cell-cycle effectors (CCND1, CDK4/6, E2F targets) and anti-apoptotic programs. Activating *ESR1* ligand-binding domain mutations (e.g., Y537S, D538G) emerging under aromatase inhibitor therapy render ERα constitutively active and drive resistance to estrogen-deprivation, yet tumors often remain ER-dependent. In EMERALD and earlier phase I data, elacestrant showed clinical activity in tumors with these *ESR1* mutations, consistent with its ability to antagonize and degrade mutant ERα. Preclinical studies demonstrated that elacestrant impedes tumor growth in ESR1-mutated, endocrine-resistant and fulvestrant-resistant PDX models, with marked suppression of estrogen-responsive transcriptional programs.

In CDK4/6 inhibitor–resistant ER+ models, elacestrant maintains activity despite alterations such as RB1 loss, CDK6 overexpression and cyclin E1/E2 upregulation, and downregulates multiple cell-cycle proteins, halting cell-cycle progression in vitro and in vivo. This indicates that ER signaling remains a key dependency even after CDK4/6i resistance, and ER degradation can re-establish control over downstream E2F and G2/M checkpoint pathways. In the clinic, EMERALD enrolled an entirely post-CDK4/6i population and still showed PFS benefit vs standard endocrine therapy, placing elacestrant as a core component of post-CDK4/6 endocrine sequencing in HR+/HER2– metastatic disease.

Primary Targets (Human)

Direct, validated human molecular targets (HGNC):

- **ESR1 (ERα)** – primary pharmacologic target. Elacestrant binds ERα, behaves as a pure antagonist, and induces proteasome-dependent ER degradation, leading to decreased estrogen-responsive gene expression in ER+ breast-cancer cells. Breast-cancer reviews and FDA labeling consistently attribute its activity to ERα targeting in ER-positive/HER2-negative, *ESR1-mutated* disease.

There is no robust human evidence that additional specific kinases or receptors are direct therapeutic targets for elacestrant beyond ERα; off-target panels have not identified a clinically relevant secondary target profile in breast cancer. Current MoA and clinical deployment are therefore best understood as *single-target ERα inhibition/degradation*.

Pathways (Overview)

Below are key pathways implicated in elacestrant biology in ER+/HER2– breast cancer, with MSigDB/Reactome/KEGG/GO identifiers where applicable. Directionality and sensitivity/resistance roles are expanded later.

1. **Estrogen response gene sets**
   - *MSigDB:* **HALLMARK_ESTROGEN_RESPONSE_EARLY**, **HALLMARK_ESTROGEN_RESPONSE_LATE**
   - *Reactome:* **R-HSA-9018519 – Estrogen-dependent gene expression**
   - *KEGG:* **hsa04915 – Estrogen signaling pathway**
     Elacestrant antagonizes and degrades ERα, leading to broad downregulation of estrogen-response gene sets in responsive PDX models and ESR1-mutant disease. Low scores for estrogen-response gene sets correlate with poor outcomes and endocrine resistance in ER+ breast cancer, highlighting these pathways as both mechanistic and predictive markers.
2. **Cell cycle & E2F targets**
   - *MSigDB:* **HALLMARK_E2F_TARGETS**, **HALLMARK_G2M_CHECKPOINT**
   - *KEGG:* **hsa04110 – Cell cycle**
     ER signaling drives cyclin D1/CDK4/6 and E2F-regulated programs. Elacestrant downregulates key cell-cycle proteins and halts cell-cycle progression in multiple ER+ models, including CDK4/6i-resistant lines, linking suppression of these pathways to drug sensitivity.
3. **PI3K/AKT/mTOR signaling**
   - *MSigDB:* **HALLMARK_PI3K_AKT_MTOR_SIGNALING**, **HALLMARK_MTORC1_SIGNALING**
   - *KEGG:* **hsa04151 – PI3K-Akt signaling pathway**, **hsa04150 – mTOR signaling pathway**
     PI3K/AKT/mTOR activation is a canonical driver of endocrine resistance (including SERDs) via ER-independent survival signaling. Preclinical work

combining elacestrant with PI3K (alpelisib) or mTOR (everolimus) inhibitors shows enhanced antitumor efficacy, implicating PI3K/AKT/mTOR downregulation as a sensitivity pathway and its activation as a resistance circuit.

4. **ESR1-mutation–driven ER signaling**
   - Functionally captured within estrogen response and ER signaling gene sets (above). Activating *ESR1* mutations maintain high estrogen-response signatures despite estrogen deprivation. EMERALD demonstrated greater PFS benefit with elacestrant in *ESR1-mutated* vs wild-type tumors, making mutant ER signaling a key *sensitivity-associated* pathway.

5. **Apoptosis pathways**
   - *MSigDB:* **HALLMARK_APOPTOSIS**
     ER blockade via SERDs promotes pro-apoptotic signaling in ER+ breast-cancer models. Elacestrant induces tumor regression in ER-dependent PDXs, consistent with engagement of apoptotic machinery downstream of ER and cell-cycle arrest, though specific apoptosis gene-set analyses for elacestrant are not as extensively reported as estrogen/cell-cycle signatures.

6. **DNA replication & repair (ER-linked)**
   - *MSigDB:* **HALLMARK_DNA_REPAIR**, **HALLMARK_MYC_TARGETS_V1**
     Estrogen signaling upregulates genes involved in proliferation and DNA replication/repair; endocrine-resistant tumors often show rewiring of these programs. While elacestrant's primary impact is via ER and cell cycle, ER-driven DNA replication and repair programs are downstream components of the same network.

7. **Endocrine-resistance meta-programs**
   Multi-omic atlases (e.g., EstroGene2.0, endocrine-resistant transcriptomic studies) identify decreased estrogen-response signatures and increased PI3K/AKT and growth-factor signaling in endocrine-relapsed tumors. These meta-programs inform resistance/sensitivity interpretation for SERDs, including elacestrant, although most analyses are class-level rather than elacestrant-specific.

Immune-specific elacestrant signatures in breast cancer have not yet been robustly defined; available data focus overwhelmingly on ER signaling and cell-cycle/PI3K networks.

Upregulated Pathways (by Context / Subtype)

Evidence-based, elacestrant-specific data on *drug-induced* transcriptional upregulation of human pathways are limited. The main mechanistic and preclinical reports (Pancholi 2022; Patel 2019; Dubash 2023) emphasize *downregulation* of ER and cell-cycle programs rather than consistent adaptive upregulated pathways.

At present, no reproducible human breast-cancer data demonstrate a particular pathway that is *consistently* upregulated on elacestrant exposure and clearly linked to either sensitivity or resistance. Therefore:

- **Elacestrant-associated upregulated pathways (human breast-cancer data):** []
  Notes: Adaptive resistance circuits in endocrine-treated ER+ disease (e.g., RTK/PI3K activation) are well described, but they have been characterized mainly for AIs and fulvestrant; specific transcriptomic "on-treatment" elacestrant upregulation signatures have not yet been published in sufficient detail.

Downregulated Pathways (by Context / Subtype)

Here we focus on pathways *experimentally shown* to be downregulated with elacestrant exposure in human-derived breast-cancer models.

1. **Estrogen response gene sets (ER+/HER2–; ESR1-WT and ESR1-mut)**
    - **Pathways:** HALLMARK_ESTROGEN_RESPONSE_EARLY/LATE; R-HSA-9018519; hsa04915
    - **Direction with elacestrant:** *Downregulated* in responding PDXs and cell models.
    - **Evidence:** Elacestrant exhibits "strong anti-estrogenic activity" in ER+ endocrine-resistant and fulvestrant-resistant PDX models, suppressing ER-dependent transcription and impeding tumor progression, particularly in *ESR1-mutated* tumors. Pharmacology reviews and PK/PD analyses confirm potent suppression of estrogen-response biomarkers with clinically relevant exposures.
    - **Clinical/subtype context:** HR+/HER2–, enriched in *ESR1-mutated* disease.
2. **Cell-cycle gene sets (HR+/HER2–, post-CDK4/6i)**
    - **Pathways:** HALLMARK_E2F_TARGETS, HALLMARK_G2M_CHECKPOINT, hsa04110

- **Direction:** *Downregulated*
- **Evidence:** In CDK4/6i-resistant ER+ models (ESR1 WT and mutant), elacestrant inhibits growth and “downregulates several key cell-cycle players” (including cyclins and CDKs), and halts cell-cycle progression both in vitro and in xenografts. This implies suppression of E2F and G2/M checkpoint pathways as part of the on-target response.

3. **ER-linked survival/proliferation meta-programs**
    - Not mapped to a single named pathway, but transcriptomic data from endocrine-responsive tumors show broad deactivation of ESR-dependent signaling and estrogen-dependent gene expression upon ER antagonism/degradation. For elacestrant, these are captured within the estrogen response and cell-cycle signatures above.

Sensitivity Pathways (by Context / Subtype)

1. Sensitivity Pathways for *Upregulation*

These are pathways where *higher baseline activity* or presence correlates with increased elacestrant benefit.

1. **ER signaling / estrogen-response activity (HR+/HER2–; *ESR1-mutated* enriched)**
    - **Pathways:** HALLMARK_ESTROGEN_RESPONSE_EARLY/LATE; R-HSA-9018519; hsa04915
    - **Rationale:**
        - Endocrine-therapy response correlates with high estrogen-response gene-set scores; endocrine-relapsed tumors often have *lower* estrogen-response scores and worse outcomes.
        - *ESR1* ligand-binding mutations (Y537S, D538G, others) maintain high ER transcriptional activity under estrogen deprivation; tumors with these mutations remained highly ER-driven in modeling studies using GSEA of estrogen-response gene sets.
        - EMERALD showed greater PFS benefit for elacestrant in *ESR1-mutated* tumors compared with wild-type, and preclinical PDX models harboring *ESR1* mutations showed strong responses to elacestrant with marked suppression of estrogen-response programs.

- **Mechanistic interpretation:** Tumors that remain strongly ER-addicted (high estrogen-response activity, especially via mutant ERα) are exquisitely sensitive to a potent oral SERD like elacestrant. Upregulation of ER signaling is thus a *sensitivity-associated* state – elacestrant removes a dominant oncogenic driver.

2. **ER-driven luminal identity (Luminal A/B)**
   - **Pathways:** Luminal ER+ transcriptional programs encompassing ESR1 target genes (overlaps with estrogen-response gene sets).
   - **Rationale:** Multi-omics compendia (e.g., EstroGene2.0) show that luminal, ER-high tumors with preserved ER transcriptional programs derive greater benefit from endocrine therapies across multiple settings. Since elacestrant's approval and guideline placement are entirely within ER+/HER2– disease, upregulated luminal/ER programs are implicitly a prerequisite for benefit.

Summary: Upregulated ER/estrogen-response signaling – particularly driven by *ESR1* mutations – is the clearest elacestrant sensitivity pathway.

2. Sensitivity Pathways for *Downregulation*

These are pathways where inhibition or suppression increases elacestrant efficacy.

1. **PI3K/AKT/mTOR signaling (HR+/HER2–; endocrine-resistant)**
   - **Pathways:** HALLMARK_PI3K_AKT_MTOR_SIGNALING, HALLMARK_MTORC1_SIGNALING; hsa04151, hsa04150
   - **Evidence (elacestrant-specific):**
     - Patel et al. tested elacestrant alone and in combination with the PI3K inhibitor alpelisib and mTOR inhibitor everolimus in multiple ER+ models (including CDK4/6i-resistant PDXs), demonstrating that combinations enhanced antitumor activity vs monotherapy.
   - **Evidence (class-level endocrine resistance):**
     - PI3K/AKT/mTOR activation is a major driver of endocrine resistance in ER+ breast cancer; AKT blockade can restore SERD responsiveness in endocrine-resistant models.
   - **Mechanistic interpretation:** Downregulation of PI3K/AKT/mTOR through targeted inhibitors removes an ER-independent survival pathway, forcing tumors to rely more heavily on ER signaling, which elacestrant then blocks. Thus, *downregulation of PI3K/AKT/mTOR enhances elacestrant sensitivity* (supported by preclinical combination data).

2. **Cell-cycle axis (CDK4/6, cyclins, E2F – HR+/HER2–, post-CDK4/6i)**
   - **Pathways:** HALLMARK_E2F_TARGETS, HALLMARK_G2M_CHECKPOINT, hsa04110
   - **Evidence:** Elacestrant itself downregulates cell-cycle proteins and halts cell-cycle progression in CDK4/6i-resistant models, indicating that tumors in which elacestrant can more completely suppress E2F/G2M programs are more sensitive.
   - **Combination-level logic:** Class-level SERD/SERM + CDK4/6 inhibitor combinations show synergistic tumor growth inhibition in endocrine-resistant models, suggesting that additional pharmacologic downregulation of the cell-cycle axis should further sensitize tumors to ER degradation.
3. **Growth-factor escape pathways (ERBB/RTK) – indirect**
   - **Evidence:** Extensive endocrine-resistance literature shows that RTK/ERBB activation (HER2, FGFR, EGFR) provides ER-independent proliferation; combined ER and RTK/PI3K blockade often restores sensitivity to hormone therapy. While elacestrant-specific RTK combination trials are early, mechanistic reasoning from SERDs suggests that suppression of RTK co-drivers would enhance elacestrant benefit.
   - **Status:** Evidence is currently class-level (SERD/endocrine) rather than elacestrant-specific; this should be treated as *hypothesis-supportive* rather than fully validated for elacestrant.

Resistance Pathways (by Context / Subtype)

3. Resistance Pathways for *Upregulation*

These are pathways whose *activation* is associated with reduced benefit from ER-targeted therapy (SERDs), and are mechanistically likely to blunt elacestrant efficacy.

1. **PI3K/AKT/mTOR hyperactivation (HR+/HER2–, especially PIK3CA-mut)**
   - **Pathways:** HALLMARK_PI3K_AKT_MTOR_SIGNALING, HALLMARK_MTORC1_SIGNALING; hsa04151, hsa04150
   - **Evidence:**
     - Genomic studies of endocrine-resistant ER+ metastatic breast cancer show frequent PIK3CA and other PI3K/AKT/mTOR pathway

alterations in tumors progressing on endocrine therapy including SERDs.

  - Preclinical work demonstrates that AKT activation diminishes SERD efficacy, and AKT inhibition resensitizes endocrine-resistant cells to SERD treatment.

- **Interpretation for elacestrant:** While EMERALD subgroup analyses suggest elacestrant retains activity in PIK3CA-mutated tumors, persistent or marked PI3K/AKT/mTOR upregulation constitutes a plausible resistance pathway, potentially limiting depth/duration of response. Evidence is strongest at the endocrine/SERD-class level.

2. **RTK/ERBB signaling upregulation (HER2, EGFR, FGFR; luminal to HER2-enriched shift)**

   - **Pathways (KEGG examples):** hsa04012 (ErbB signaling), hsa04010 (MAPK signaling)

   - **Evidence:** Endocrine-resistant ER+ tumors often show increased RTK signaling and HER2-enriched transcriptomic features, which correlate with reduced sensitivity to endocrine therapy and fulvestrant.

   - **Relevance to elacestrant:** No direct clinical data yet link RTK upregulation to elacestrant resistance, but mechanistic logic suggests that tumors relying more on ERBB/RTK pathways and less on ER would be less responsive to ER degradation alone. This is best viewed as a *class-level SERD resistance circuit*.

3. **General cell-cycle deregulation to ER independence**

   - **Evidence:** CDK4/6i-resistant models can evolve ER-independent proliferation via CDK6 amplification or cyclin E overexpression; some of these are less dependent on ER signaling. However, Patel et al. specifically found that many CDK4/6i-resistant models remained sensitive to elacestrant despite such changes.

   - **Interpretation:** While extreme ER-independent cell-cycle activation could theoretically mediate resistance, current elacestrant data actually show preserved sensitivity across diverse cell-cycle resistance mechanisms; at present there is *no specific cell-cycle upregulation pattern clearly linked to elacestrant resistance*.

4. Resistance Pathways for *Downregulation*

These are pathways where loss or suppression undermines elacestrant activity.

1. **Loss/downregulation of ER expression and estrogen-response programs**
    - **Pathways:** HALLMARK_ESTROGEN_RESPONSE_EARLY/LATE; R-HSA-9018519
    - **Evidence:**
        - Transcriptomic analyses of endocrine-relapsed ER+ tumors often show downregulation of early and late estrogen-response gene sets, ESR-dependent signaling and estrogen-dependent gene expression, with low estrogen-response scores predicting worse prognosis and endocrine resistance.
        - Clinical endocrine-therapy resistance includes ER loss or marked ER downregulation, which renders SERMs/SERDs ineffective due to lack of target.
    - **Implication for elacestrant:** Elacestrant requires ERα for binding and degradation. Tumors that have evolved ER loss or very low estrogen-response activity are intrinsically resistant, as seen for endocrine agents in general. EMERALD's stronger benefit in *ESR1-mutated* vs broader ER+ cohorts is consistent with poorer relative efficacy in tumors with weaker ER dependence.
2. **Apoptotic pathway suppression (class-level)**
    - **Evidence:** General endocrine resistance is associated with reduced pro-apoptotic signaling and increased survival pathways. However, there are no elacestrant-specific datasets directly tying low apoptosis gene-set activity to resistance.
    - **Status:** Potential class-level SERD resistance mechanism; currently insufficient elacestrant-specific human data → *treat as hypothesis-generating rather than validated*.

Because of strict evidence requirements, no additional elacestrant-specific resistance pathways can be confidently assigned beyond ER loss/low estrogen-response signatures and PI3K/AKT/mTOR/RTK activation as class-level endocrine resistance circuits.

Breast Cancer Subtype–Stratified Evidence

- **HR+/HER2– (Luminal A/B)**
    - *All* robust elacestrant clinical data (phase I, EMERALD) and approvals are in ER+/HER2– advanced or metastatic disease.

- EMERALD: post-CDK4/6i population; elacestrant vs fulvestrant/AI, with PFS benefit overall and particularly in *ESR1-mutated* tumors.
- Preclinical PDX models encompass AI-resistant, fulvestrant-resistant, CDK4/6i-resistant ER+ tumors and support efficacy across ESR1 WT and mutant settings.

- **HER2-positive**
  - No approved indications or phase III trials of elacestrant in HER2-positive disease. HER2-amplified ER+ tumors may have reduced ER-dependence and greater reliance on ERBB signaling; endocrine/SERD benefit is context-dependent and generally secondary to anti-HER2 therapy.
  - **Evidence for elacestrant:** insufficient → no validated sensitivity or resistance mechanisms specific to HER2-positive breast cancer.
- **Triple-negative breast cancer (TNBC)**
  - TNBC is generally ER-negative; ER is absent or negligible, eliminating the pharmacologic target for elacestrant.
  - No meaningful clinical or preclinical evidence supports elacestrant use in TNBC; any observed ER expression in a “TNBC” label would likely reclassify the tumor biologically as ER+.
- **gBRCA1/2-mutated / HRD+**
  - While gBRCA/HRD status strongly predicts PARP inhibitor benefit, there is no evidence that it directly modulates elacestrant sensitivity beyond its association with luminal biology in some ER+ tumors. No specific elacestrant–BRCA/HRD interaction has been established.

Contraindications and Safety

Formal Contraindications (US FDA label)

- **Contraindications:** *None.*

Key Warnings and Precautions

1. **Dyslipidemia**
   - Hypercholesterolemia (any grade ~30%) and hypertriglyceridemia (~27%) were common in EMERALD; grade 3–4 events were infrequent but present.

    - *Recommendation:* Monitor lipid profile prior to and periodically during therapy.

2. **Embryo-fetal toxicity**
    - Based on animal data and MoA, elacestrant can cause embryo-fetal harm; embryo-fetal mortality and structural abnormalities occurred in rats at exposures below the clinical AUC.
    - *Recommendation:*
        - Avoid use in pregnancy; verify pregnancy status prior to therapy.
        - Use effective contraception during treatment and for 1 week after last dose (both sexes).

3. **Hepatic impairment**
    - Avoid use in **severe hepatic impairment (Child-Pugh C)**; dose reduction required for **moderate impairment (Child-Pugh B)**; no adjustment for mild impairment.

4. **Drug–drug interactions (CYP3A4, P-gp, BCRP)**
    - Elacestrant is metabolized by CYP3A4; strong/moderate CYP3A4 inducers or inhibitors should be *avoided* due to risk of decreased efficacy or increased toxicity.
    - Elacestrant is an inhibitor of P-gp and BCRP; coadministration with substrates can raise substrate levels and toxicities.

5. **Common adverse reactions (≥10%)**
    - Musculoskeletal pain, nausea, vomiting, fatigue, headache, hot flush, abdominal pain, dyspepsia; lab abnormalities include increased cholesterol, triglycerides, AST/ALT, creatinine and decreased hemoglobin, sodium.
    - Overall serious adverse events were ~12% in EMERALD; fatal events ~1.7%.

Guideline-based "avoid / not recommended" contexts

- All major guidelines restrict elacestrant use to **ER+/HER2–, *ESR1-mutated* advanced/metastatic breast cancer after prior endocrine therapy including CDK4/6 inhibitor**, generally after ≥1 line of therapy in the metastatic setting.
- Not recommended in ER-negative disease or outside an ESR1-mutated context, except in clinical trials.

Trial and Guideline Context

Key clinical trials

- **Phase I trial (Bardia et al., JCO 2021; NCT02338349)**
  - ER+/HER2– advanced breast cancer, heavily pretreated.
  - Established recommended dose (345 mg once daily); observed partial responses and durable stable disease, including in *ESR1-mutated* tumors.
- **EMERALD (Phase III; NCT03778931)**
  - Population: ER+/HER2– advanced/metastatic breast cancer, all post-CDK4/6 inhibitor and ≥1 prior endocrine regimen.
  - Design: Elacestrant 345 mg daily vs physician's-choice endocrine therapy (fulvestrant or AI).
  - Outcomes:
    - **Overall population:** improved PFS (HR ~0.70) vs control.
    - **ESR1-mutated subgroup:** larger PFS benefit (HR ~0.55; median PFS ~3.8 vs 1.9 months).
  - Toxicities were predominantly GI and dyslipidemia, generally manageable.
- **Ongoing/early-phase combination trials**
  - Trials are exploring elacestrant with CDK4/6 inhibitors and PI3K/mTOR inhibitors, including in patients with brain metastases and in earlier-line settings. Human mechanistic data are still emerging; current combination rationale is largely supported by preclinical synergy with alpelisib/everolimus.

Guideline placement

- **NCCN (Breast Cancer Guidelines)**
  - Lists elacestrant as an oral SERD for ER+/HER2– disease, and specifically as a treatment option for **ESR1-mutated** metastatic breast cancer after progression on endocrine therapy plus CDK4/6 inhibitor.
- **ESMO Living Guideline – HR+/HER2– MBC**

- Recommends elacestrant for patients with HR+/HER2–, *ESR1-mutated* advanced breast cancer, scoring it MCBS 3 and ESCAT I-A for ESR1 mutation as the targetable biomarker.

- **ASCO rapid recommendation (ESR1 testing & therapy)**
  - Advises routine *ESR1* mutation testing at progression on endocrine therapy in HR+/HER2– MBC and lists elacestrant as a treatment option for patients with detectable *ESR1* mutations after prior CDK4/6 inhibitor.

Collectively, these place elacestrant as a post-CDK4/6, ESR1-mutation–directed oral SERD with a clear biomarker-defined niche in HR+/HER2– metastatic breast cancer.

Pathway Evidence Table

Each row summarizes a key pathway, regulation context, and its effect on elacestrant sensitivity or resistance. Where evidence is class-level (SERD/endocrine) rather than elacestrant-specific, this is noted in the rationale.

Pathway ID / Name	Regulation (Up/Down)	Effect of Drug (Sensitive / Resistant)	Biological Rationale (breast-cancer specific)	Key Reference(s)
HALLMARK_ESTROGEN_RESPONSE_EARLY / **HALLMARK_ESTROGEN_RESPONSE_LATE**	**Down** (on elacestrant treatment)	**Sensitive**	In ER+/HER2– PDX models, elacestrant shows strong anti-estrogenic activity, suppressing ER-dependent gene expression and impeding tumor progression, especially in *ESR1-mutated*	Pancholi 2022 (NPJ Breast Cancer); Beumer 2023 PK/PD

Pathway ID / Name	Regulation (Up/Down)	Effect of Drug (Sensitive / Resistant)	Biological Rationale (breast-cancer specific)	Key Reference (s)
			endocrine-resistant and fulvestrant-resistant tumors. Downregulation of estrogen-response gene sets marks effective target engagement and sensitivity.	
R-HSA-9018519 – Estrogen-dependent gene expression	**Down** (on treatment)	**Sensitive**	Reactome's estrogen-dependent gene-expression pathway encompasses canonical ER target genes. Elacestrant's SERD activity reduces ER protein and ER-regulated transcription, fitting suppression	Wardell 2015 (RAD1901 ER degradation); Pancholi 2022

Pathway ID / Name	Regulation (Up/Down)	Effect of Drug (Sensitive / Resistant)	Biological Rationale (breast-cancer specific)	Key Reference (s)
			of this pathway in responsive ER+/HER2– tumors.	
hsa04915 – Estrogen signaling pathway (KEGG)	**Up (baseline)**	**Sensitive**	High baseline estrogen-signaling activity (often maintained by *ESR1* mutations) is associated with endocrine responsiveness. EMERALD showed greater PFS gain from elacestrant in *ESR1-mutated* tumors, which typically preserve strong ER signaling despite AIs, making them more dependent on hsa04915 and thus more	Bidard 2022 EMERALD; Harrod 2022 ESR1-mutation GSEA; Valenza 2024 review

Pathway ID / Name	Regulation (Up/Down)	Effect of Drug (Sensitive / Resistant)	Biological Rationale (breast-cancer specific)	Key Reference (s)
			sensitive to SERD-mediated blockade.	
HALLMARK_ESTROGEN_RESPONSE_EARLY/LATE	**Down (baseline in relapsed disease)**	**Resistant**	Transcriptomic profiling of endocrine-relapsed ER+ tumors shows downregulation of early and late estrogen-response gene sets and ESR-dependent signaling. Low estrogen-response scores correlate with poor endocrine outcomes, implying that tumors with weak ER-program activity are intrinsically less responsive to SERDs,	Schagerholm Stanev 2025 (BMC Cancer); EstroGene 2.0

Pathway ID / Name	Regulation (Up/Down)	Effect of Drug (Sensitive / Resistant)	Biological Rationale (breast-cancer specific)	Key Reference (s)
			including elacestrant.	
HALLMARK_E2F_TARGETS	**Down** (on elacestrant)	**Sensitive**	In CDK4/6i-resistant ER+ models, elacestrant downregulates key cell-cycle proteins and halts cell-cycle progression, implying suppression of E2F-regulated transcription. Tumors in which E2F targets are effectively suppressed by elacestrant show strong growth inhibition, indicating a sensitivity-associated downregulation.	Patel 2019 (CDK4/6i-resistant models)

Pathway ID / Name	Regulation (Up/Down)	Effect of Drug (Sensitive / Resistant)	Biological Rationale (breast-cancer specific)	Key Reference (s)
HALLMARK_G2M_CHECKPOINT / hsa04110 – Cell cycle	**Down** (on elacestrant)	**Sensitive**	Elacestrant induces cell-cycle arrest in multiple ER+ models, including those with diverse CDK4/6i resistance mechanisms. Suppression of G2/M checkpoint and broader cell-cycle gene sets is part of the anti-proliferative response and correlates with sensitivity.	Patel 2019; Bihani 2017 (ER+ PDXs)
HALLMARK_PI3K_AKT_MTOR_SIGNALING / HALLMARK_MTORC1_SIGNALING	**Down** (via PI3K/mTOR inhibition)	**Sensitive**	Elacestrant combined with alpelisib (PI3K inhibitor) or everolimus (mTOR inhibitor) showed	Patel 2019; Ribas 2015 (AKT antagonism + SERD)

Pathway ID / Name	Regulation (Up/Down)	Effect of Drug (Sensitive / Resistant)	Biological Rationale (breast-cancer specific)	Key Reference (s)
			enhanced antitumor activity in ER+ and CDK4/6i-resistant models, consistent with downregulation of PI3K/AKT/mTOR as a sensitizing mechanism that removes ER-independent survival signals.	
HALLMARK_PI3K_AKT_MTOR_SIGNALING / hsa04151 – PI3K-Akt signaling	**Up** (baseline or acquired)	**Resistant**	Genomic and expression analyses of endocrine-resistant ER+ metastatic breast cancer show frequent PIK3CA and PI3K/AKT/mTOR pathway activation in tumors progressing	Razavi 2018 (endocrine-resistant ER+ MBC); endocrine-resistance reviews

Pathway ID / Name	Regulation (Up/Down)	Effect of Drug (Sensitive / Resistant)	Biological Rationale (breast-cancer specific)	Key Reference (s)
			on endocrine therapy including SERDs, correlating with endocrine resistance. Persistent activation likely diminishes elacestrant benefit, although EMERALD suggests some retained activity in PIK3CA-mut tumors.	
ErbB/RTK signaling (e.g., hsa04012 – ErbB signaling pathway)	**Up**	**Resistant**	ER+ tumors progressing on endocrine therapy often exhibit increased HER2/EGFR/FGFR signaling, with HER2-enriched phenotypes showing less	Endocrine-resistance reviews (e.g., Nunnery 2020, summarized in Bardia 2024)

Pathway ID / Name	Regulation (Up/Down)	Effect of Drug (Sensitive / Resistant)	Biological Rationale (breast-cancer specific)	Key Reference (s)
			benefit from ER-targeted therapy. Upregulated RTK signaling can bypass ER dependency, and although elacestrant-specific data are lacking, this is a recognized class-level resistance circuit for SERDs.	
Apoptosis pathways (HALLMARK_APOPTOSIS)	**Down**	**Resistant**	Diminished pro-apoptotic signaling is a general hallmark of endocrine resistance. SERD-treated models that fail to activate apoptosis despite ER suppression tend to show incomplete responses.	ER-signaling and endocrine-resistance overviews

Pathway ID / Name	Regulation (Up/Down)	Effect of Drug (Sensitive / Resistant)	Biological Rationale (breast-cancer specific)	Key Reference (s)
			For elacestrant, this is inferred from class-level data; direct pathway-level human datasets are still limited.	

Note: Where data are not explicitly elacestrant-specific (e.g., RTK and some PI3K/AKT/mTOR resistance and apoptosis pathways), the table reflects *SERD/endocrine-class evidence in ER+/HER2– breast cancer* with a mechanistic rationale for relevance to elacestrant; such entries should be interpreted as supported but not yet fully validated for elacestrant alone.

Citations

Bardia, A., et al. (2021). Phase I study of elacestrant (RAD1901) in ER-positive, HER2-negative advanced breast cancer. *Journal of Clinical Oncology, 39*(12), 1360–1370. PMID: 33513026.

Bidard, F.-C., et al. (2022). Elacestrant versus standard endocrine monotherapy for ER-positive, HER2-negative metastatic breast cancer (EMERALD): a phase 3 randomized trial. *Journal of Clinical Oncology*. PMID: 35584336. NCT03778931.

Bardia, A., et al. (2024). Elacestrant in ER-positive, HER2-negative metastatic breast cancer with ESR1 mutations: Subgroup analyses from EMERALD. *Clinical Cancer Research, 30*(19), 4299–4312. PMID: 39087959.

Bihani, T., et al. (2017). Elacestrant (RAD1901), a selective estrogen receptor degrader, has antitumor activity in multiple ER-positive breast cancer models. *Clinical Cancer Research, 23*(16), 4793–4804. PMID: 28473534.

Patel, H. K., et al. (2019). Elacestrant (RAD1901) demonstrates antitumor activity in CDK4/6 inhibitor–resistant ER-positive breast cancer models. *Breast Cancer Research, 21*, 146. PMID: 31801598.

Pancholi, S., et al. (2022). Elacestrant shows strong anti-estrogenic activity in endocrine-resistant and fulvestrant-resistant ER-positive breast cancer patient-derived xenografts. *NPJ Breast Cancer, 8*, 121. PMID: 36317934.

Rani, A., et al. (2019). Endocrine resistance in hormone receptor-positive breast cancer: An overview. *Frontiers in Endocrinology, 10*, 245. PMID: 31133803.

Li, Z., et al. (2024). The EstroGene2.0 database for endocrine therapy response and resistance in breast cancer. *bioRxiv*. https://doi.org/10.1101/2024.06.28.601163.

Clusan, L., et al. (2023). A basic review on estrogen receptor signaling pathways in breast cancer. *International Journal of Molecular Sciences, 24*(7), 6834. PMID: 37048255.

Nath, A., et al. (2022). ENDORSE: A prognostic model for endocrine therapy in ER-positive breast cancer. *Molecular Systems Biology, 18*(4), e10558. PMID: 35318601.

Hoy, S. M. (2023). Elacestrant: First approval. *Drugs, 83*(7), 633–639. PMID: 36849468.

Varella, L., et al. (2023). Evaluating elacestrant in the management of ER-positive, HER2-negative metastatic breast cancer. *Cancers (Basel), 15*(6), 1779. PMID: 36901578.

Shah, M., et al. (2024). Elacestrant for estrogen receptor–positive, human epidermal growth factor receptor 2–negative advanced breast cancer. *The Oncologist, 29*(3), e239–e248. PMID: 38376373.

Hageman, E., et al. (2024). Elacestrant for ER-positive, HER2-negative advanced breast cancer with ESR1 mutations. *Annals of Pharmacotherapy, 58*(11), 1193–1205. PMID: 37994211.

Gheysen, M., et al. (2024). Oral SERDs changing the landscape in hormone receptor–positive metastatic breast cancer. *Cancer Treatment Reviews*. PMID: 39227547.

Fan, Z., et al. (2025). Post-marketing safety profile of elacestrant in breast cancer. *BMC Pharmacology and Toxicology, 26*(x). PMID: 39929820.

Ribas, R., et al. (2015). AKT antagonist AZD5363 influences estrogen receptor function in endocrine-resistant breast cancer and synergizes with fulvestrant in vivo. *Molecular Cancer Therapeutics, 14*(9), 2035–2048. PMID: 26206984.

Palaniappan, M., et al. (2024). Current therapeutic opportunities for estrogen receptor alpha–positive breast cancer. *Biomedicines, 12*(12), 2700. PMID: 39689175.